Probability and Random Processes

Random walks. Exercises

1. Two players, A and B, are engaged in a game. In each round, player A wins probability $p = 1/2$, receiving one unit from B, or loses with probability $q = 1/2$, paying one unit to B. Let X_n represent the net gain of A after the n-th round, where $X_0 = 0$.

(a) What is the probability of a draw in the n-th round?

(b) What is the expected number of draws in the first 6 games? [Hint: You may consider the indicator function I_k of a draw in the k-th game.]

(c) Provide an estimation of the probability that A has a net gain of at least 20 units in the 100-th round.

Solution

The stochastic process $\{X_n : n \geqslant 0\}$ is a symmetric random walk.

(a) The probability of a draw in the n-th round is given by $\mathbb{P}(X_n = 0)$. This probability equals zero if n is odd, and if $n = 2k$ is even, then

$$\mathbb{P}(X_{2k} = 0) = \binom{2k}{k}\left(\frac{1}{2}\right)^{2k}, \quad k \geqslant 1.$$

(b) Let I_k be the indicator random variable of a draw in the k–th game. Then, the number N of draws in the first 6 games is $N = I_2 + I_4 + I_6$. Therefore,

$$\begin{aligned}\mathbb{E}(N) &= \mathbb{E}(I_2) + \mathbb{E}(I_4) + \mathbb{E}(I_6) = \mathbb{P}(X_2 = 0) + \mathbb{P}(X_4 = 0) + \mathbb{P}(X_6 = 0)\\ &= \binom{2}{1}\left(\frac{1}{2}\right)^2 + \binom{4}{2}\left(\frac{1}{2}\right)^4 + \binom{6}{3}\left(\frac{1}{2}\right)^6 = \frac{2}{4} + \frac{6}{16} + \frac{20}{64} = \frac{19}{16}.\end{aligned}$$

(c) By the Central Limit Theorem, we know that $X_n/\sqrt{n}$ converges in distribution to the $\mathsf{N}(0,1)$ law as $n \to \infty$. Hence,

$$P(X_{100} \geqslant 20) = \mathbb{P}\left(\frac{X_{100}}{10} \geqslant 2\right) \approx 1 - F_{N(0,1)}(2) \approx 0.0228.$$

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2. Two players, A and B, participate in a game where they toss a fair coin. If the result is heads, player A pays one unit to player B. Conversely, if the result is tails, player B pays one unit to player A. Let X_n represent the net gain of player A after the first n coin tosses.

(a) Compute $\mathbb{P}(X_3 = -1, X_5 = -3, X_9 = -3)$.

(b) If $X_{50} = 0$, what is the probability that $\mathbb{P}(X_{53} = -1, X_{55} = -3, X_{59} = -3)$.

(c) By using the Central Limit Theorem, give an approximate value for $\mathbb{P}(-196 \leqslant X_{10000} \leqslant 196)$.

Solution

The process $\{X_n : n \geqslant 0\}$ is a symmetric random walk.

(a) We have

$$\mathbb{P}(X_3 = -1, X_5 = -3, X_9 = -3) = \mathbb{P}(X_3 - X_0 = -1, X_5 - X_3 = -2, X_9 - X_5) = 0).$$

Since the increments $X_3 - X_0$, $X_5 - X_3$, and $X_9 - X_5$ are independent random variables, and, if $n > m$, $\mathbb{P}(X_n - X_m = a) = \mathbb{P}(X_{n-m} = a)$, the sought probability equals

$$\mathbb{P}(X_3 = -1)\,\mathbb{P}(X_2 = -2)\,\mathbb{P}(X_4 = 0).$$

Now, by using the first-order probability function of a symmetric random walk,

$$\mathbb{P}(X_n = k) = \binom{n}{\frac{n+k}{2}} \left(\frac{1}{2}\right)^n,$$

we get

$$\mathbb{P}(X_3 = -1)\,\mathbb{P}(X_2 = -2)\,\mathbb{P}(X_4 = 0) = \binom{3}{1}\left(\frac{1}{2}\right)^3 \binom{2}{0}\left(\frac{1}{2}\right)^2 \binom{4}{2}\left(\frac{1}{2}\right)^4 = \frac{9}{256} \approx 0{,}0352.$$

(b) By taking into account that the random walk has independent increments, and that it is temporally homogeneous, the probability $\mathbb{P}(X_{53} = -1, X_{55} = -3, X_{59} = -3)$, if $X_{50} = 0$, is given by:

$$\begin{aligned}
&\mathbb{P}(X_{53} = -1, X_{55} = -3, X_{59} = -3 \,|\, X_{50} = 0) \\
&\quad = \frac{\mathbb{P}(X_{50} = 0, X_{53} = -1, X_{55} = -3, X_{59} = -3)}{\mathbb{P}(X_{50} = 0)} \\
&\quad = \frac{\mathbb{P}(X_{50} = 0, X_{53} - X_{50} = -1, X_{55} - X_{53} = -2, X_{59} - X_{55} = 0)}{\mathbb{P}(X_{50} = 0)} \\
&\quad = \frac{\mathbb{P}(X_{50} = 0)\,\mathbb{P}(X_{53} - X_{50} = -1)\,\mathbb{P}(X_{55} - X_{53} = -2)\,\mathbb{P}(X_{59} - X_{55} = 0)}{\mathbb{P}(X_{50} = 0)} \\
&\quad = \mathbb{P}(X_{53} - X_{50} = -1)\,\mathbb{P}(X_{55} - X_{53} = -2)\,\mathbb{P}(X_{59} - X_{55} = 0) \\
&\quad = \mathbb{P}(X_3 = -1)\mathbb{P}(X_2 = -2)\mathbb{P}(X_4 = 0),
\end{aligned}$$

where the last probability is computed in part (a).

(c) We know that the random variable X_n can be expressed as $Y_1 + Y_2 + \cdots + Y_n$, where the variables Y_i are independent and identically distributed, such that $\mathbb{P}(Y_i = -1) = \mathbb{P}(Y_i = 1) = 1/2$. We have $\mathbb{E}(Y_i) = 0$ and $\mathrm{Var}(Y_i) = 1$, and the Central Limit Theorem states that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{Y_i - \mathbb{E}(Y_i)}{\sqrt{\mathrm{Var}(Y_i)}} = \frac{X_n}{\sqrt{n}}$$

converges in distribution to the standard normal distribution, as $n \to \infty$. Thus, the desired probability can be approximated as follows:

$$\mathbb{P}(-196 \leqslant X_{10000} \leqslant 196) = \mathbb{P}\left(-1.96 \leqslant \frac{X_{10000}}{100} \leqslant 1.96\right) = F_{N(0,1)}(1.96) - F_{N(0,1)}(-1.96) \approx 0.950.$$

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3. Let $\{X_n : n \geqslant 0\}$ be a random walk with parameter $p = 3/4$.

(a) What is the probability of a return to the origin at time k?

(b) If at time n we have $X_n = a$, what is the probability of visiting again a after k steps?

(c) Calculate the expected number of visits to the origin during the interval $1 \leqslant n \leqslant 6$. What is this number if $p = 1/4$?

Solution

(a) The probability of a return to the origin at time k is 0 if k is odd, and if k is even, then

$$\mathbb{P}(X_k = 0) = \binom{k}{\frac{k}{2}} p^{k/2} q^{k/2}.$$

(b) Since the steps of a random walk are independent, once we reach position a at time n, we have

$$\mathbb{P}(X_{n+k} = b \,|\, X_n = a) = \mathbb{P}(X_k = b \,|\, X_0 = a) = \mathbb{P}(X_k = b - a \,|\, X_0 = 0) = \mathbb{P}(X_k = b - a).$$

In particular,

$$\mathbb{P}(X_{n+k} = a \,|\, X_n = a) = \mathbb{P}(X_k = 0),$$

which has been computed in part (a).

(c) Let I_k be the indicator random variable for the event "the walk visits the origin at time k", so that $\mathbb{E}(I_k) = \mathbb{P}(X_k = 0)$. Since this probability is zero when k is odd, we can conclude that the expected number of visits to the origin, for $1 \leqslant k \leqslant 6$, is given by:

$$\begin{aligned}\mathbb{E}\left(\sum_{i=1}^{6} I_k\right) &= \sum_{i=1}^{6} \mathbb{E}(I_i) = \sum_{i=1}^{6} \mathbb{P}(X_i = 0) \\ &= \mathbb{P}(X_2 = 0) + \mathbb{P}(X_4 = 0) + \mathbb{P}(X_6 = 0) = \binom{2}{1} pq + \binom{4}{2} p^2q^2 + \binom{6}{3} p^3q^3.\end{aligned}$$

In particular, if $p = 3/4$, we get:

$$\mathbb{E}\left(\sum_{i=1}^{6} I_k\right) = 2 \cdot \frac{3}{4^2} + 6 \cdot \frac{9}{4^4} + 20 \cdot \frac{27}{4^6} = \frac{735}{1024} \approx 0.7178.$$

By symmetry, we obtain the same value for $p = 1/4$.

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4. Let $\{X_n : n \geqslant 0\}$ be a symmetric random walk. Calculate the following probabilities:

(a) That the random walk does not visit the coordinate -1 for any $1 \leqslant n \leqslant 2m$.

(b) That the random walk does not visit the coordinate -2 for any $1 \leqslant n \leqslant 2m$.

(c) That $X_n \leqslant 2$ for all $1 \leqslant n \leqslant 2m$,

Solution

(a) The probability α_m that the random walk does not visit the coordinate -1 during the first $2m$ steps can be expressed as:

$$\alpha_m = \mathbb{P}(X_1 \geqslant 0, \ldots, X_{2m} \geqslant 0).$$

Let us prove that α_m is also equal to $\mathbb{P}(X_1 \neq 0, \ldots, X_{2m} \neq 0)$. To see this, consider the following facts:

(a.1) If $X_{2m-1} \geqslant 0$, then $X_{2m-1} > 0$, and so $X_{2m} \geqslant 0$. Therefore,

$$\mathbb{P}(X_1 \geqslant 0, \ldots, X_{2m-1} \geqslant 0, X_{2m} \geqslant 0) = \mathbb{P}(X_1 \geqslant 0, \ldots, X_{2m-1} \geqslant 0).$$

(a.2) By the homogeneity of the random walk, and taking into account that $X_0 = 0$,

$$\begin{aligned}\mathbb{P}(X_1 \geqslant 0, \ldots, X_{2m-1} \geqslant 0) &= \mathbb{P}(X_1 \geqslant 0, \ldots, X_{2m-1} \geqslant 0 \,|\, X_0 = 0) \\ &= \mathbb{P}(X_2 \geqslant 1, \ldots, X_{2m} \geqslant 1 \,|\, X_1 = 1) \\ &= \mathbb{P}(X_2 > 0, \ldots, X_{2m} > 0 \,|\, X_1 = 1).\end{aligned}$$

(a.3) Moreover,

$$\begin{aligned}\mathbb{P}(X_2 > 0, \ldots, X_{2m} > 0 \,|\, X_1 = 1) &= \frac{\mathbb{P}(X_1 = 1, X_2 > 0, \ldots, X_{2m} > 0)}{\mathbb{P}(X_1 = 1)} \\ &= 2\,\mathbb{P}(X_1 > 0, X_2 > 0, \ldots, X_{2m} > 0).\end{aligned}$$

(a.4) Furthermore, by the symmetry of the random walk,

$$\mathbb{P}(X_1 > 0, \ldots, X_{2m} > 0) = \mathbb{P}(X_1 < 0, \ldots, X_{2m} < 0).$$

Hence,

$$\begin{aligned}2\,\mathbb{P}(X_1 > 0, \ldots, X_{2m} > 0) &= \mathbb{P}(X_1 > 0, \ldots, X_{2m} > 0) + \mathbb{P}(X_1 < 0, \ldots, X_{2m} < 0) \\ &= \mathbb{P}(X_1 \neq 0, \ldots, X_{2m} \neq 0).\end{aligned}$$

Then, putting all together,

$$\begin{aligned}\alpha_m &= \mathbb{P}(X_1 \geqslant 0, \dots, X_{2m} \geqslant 0) = \mathbb{P}(X_1 \geqslant 0, \dots, X_{2m-1} \geqslant 0)\\ &= \mathbb{P}(X_2 > 0, \dots, X_{2m} > 0 \,|\, X_1 = 1) = 2\,\mathbb{P}(X_1 > 0, X_2 > 0, \dots, X_{2m} > 0)\\ &= \mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0),\end{aligned}$$

as we wanted to prove.
Finally, by using the known fact that $\mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0) = \mathbb{P}(X_{2m} = 0)$ we get

$$\alpha_m = \binom{2m}{m}\frac{1}{2^{2m}}.$$

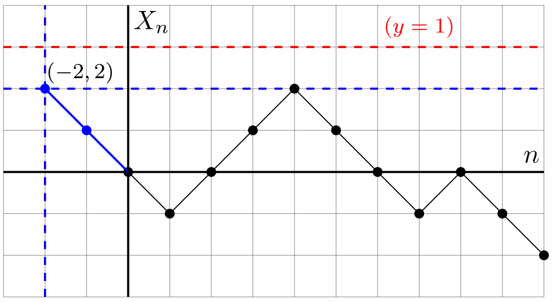
(b) By the spatial and temporal homogeneity of the random walk, and taking into account that if $X_{2m+1} \geqslant 0$, then $X_{2m+2} \geqslant 0$, we deduce (reasoning as in the previous item) that:

$$\begin{aligned}\mathbb{P}(X_1 \geqslant -1, \dots, X_{2m} \geqslant -1) &= \mathbb{P}(X_2 \geqslant 0, \dots, X_{2m+1} \geqslant 0 \,|\, X_1 = 1)\\ &= 2\,\mathbb{P}(X_1 \geqslant 0, X_2 \geqslant 0, \dots, X_{2m+1} \geqslant 0)\\ &= 2\,\mathbb{P}(X_1 \geqslant 0, X_2 \geqslant 0, \dots, X_{2m+1} \geqslant 0, X_{2m+2} \geqslant 0)\\ &= 2\,\alpha_{m+1} = \binom{2m+2}{m+1}\frac{1}{2^{2m+1}} = \binom{2m+1}{m}\frac{1}{2^{2m}},\end{aligned}$$

where in the last equality, we have used that

$$\binom{2m+2}{m+1} = \binom{2m+1}{m} + \binom{2m+1}{m+1} = \binom{2m+1}{m} + \binom{2m+1}{m} = 2\binom{2m+1}{m}$$

(c) Let us consider the (n, X_n)-plane representation of the random walk. By placing the origin at $(-2, 2)$, we observe that the probability that $X_n \leqslant 2$ for $n = 1, 2, \dots, 2m$ corresponds to the probability that a path of length $2m+2$ does not touch the line $y = 1$, given that its first two steps are both "down".



In other words, if $X_1' = -1$, $X_2' = -2$, and $X_n' = X_{n-2} - 2$ for $n \geqslant 3$, then

$$\begin{aligned}\mathbb{P}(X_1 \leqslant 2, \dots, X_{2m} \leqslant 2) &= \mathbb{P}(X_1' = -1, X_2' = -2, X_3' \leqslant 0, \dots, X_{2m+2}' \leqslant 0 \,|\, X_1' = -1, X_2' = -2)\\ &= 4\,\mathbb{P}(X_1' = -1, X_2' = -2, X_3' \leqslant 0, \dots, X_{2m+2}' \leqslant 0)\\ &= 4\,\mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0),\end{aligned} \tag{1}$$

where the last equality is because $\{X_n : n \geqslant 1\}$ and $\{X_n' : n \geqslant 1\}$ are both symmetric random walks. Moreover, notice that if α_m is defined as in part (a), by the symmetry of the random walk, we have

$$\begin{aligned}\alpha_{m+1} &= \mathbb{P}(X_1 \leqslant 0, X_2 \leqslant 0, \dots, X_{2m+2} \leqslant 0)\\ &= \mathbb{P}(X_1 = -1, X_2 = 0, X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0) + \mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0).\end{aligned}$$

Thus, since

$$\begin{aligned}&\mathbb{P}(X_1 = -1, X_2 = 0, X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0)\\ &\quad= \mathbb{P}(X_1 = -1, X_2 = 0)\,\mathbb{P}(X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0 \,|\, X_2 = 0, X_1 = 0)\\ &\quad= \frac{1}{4}\,\mathbb{P}(X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0 \,|\, X_2 = 0)\\ &\quad= \frac{1}{4}\,\mathbb{P}(X_1 \leqslant 0, \dots, X_{2m} \leqslant 0 \,|\, X_0 = 0) = \frac{1}{4}\,\mathbb{P}(X_1 \leqslant 0, \dots, X_{2m} \leqslant 0)\\ &\quad= \frac{1}{4}\alpha_m,\end{aligned}$$

and, by (1),

$$\mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leqslant 0, \dots, X_{2m+2} \leqslant 0) = \frac{1}{4}\,\mathbb{P}(X_1 \leqslant 2, \dots, X_{2m} \leqslant 2),$$

we can conclude that

$$\alpha_{m+1} = \frac{1}{4}\alpha_m + \frac{1}{4}\,\mathbb{P}(X_1 \leqslant 2, \dots, X_{2m} \leqslant 2).$$

Finally, we obtain that:

$$\begin{aligned}\mathbb{P}(X_1 \leqslant 2, \dots, X_{2m} \leqslant 2) &= 4\alpha_{m+1} - \alpha_m \\ &= \left(\binom{2m+2}{m+1} - \binom{2m}{m}\right)\frac{1}{2^{2m}} = \frac{3m+1}{m+1}\binom{2m}{m}\frac{1}{2^{2m}}.\end{aligned}$$

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5. A coin is tossed $n = 20$ times. Let Y be the random variable that counts the last time that the number of heads is equal to the number of tails ($Y = 0$ means that the game is not in a draw during the 20 tosses).

(a) Calculate $\mathbb{P}(Y = 2k)$.

(b) What is more probable, that the game has not been in a draw or that the last draw were for $k = 5$.

Solution

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6. In an election, there are two candidates, A and B, and each of $2m$ voters votes with equal probability for A or B.

(a) If candidate A has received two votes more than candidate B, what is the probability that A has always been in the lead during the counting of the votes?

(b) What is the answer to the previous question if A has received $2r$ votes more than B?

(c) What is the probability that one of the candidates has always been in the lead while the votes are being counted?

(d) If candidate A has never fallen behind candidate B during the counting of the votes, calculate the probability that the counting finishes with an equal number of votes for both candidates.

Solution

Each counting of the votes can be understood as a path in the (n, X_n)-plane represenation, where a step "up" represents a vote for candidate A, while a step "down" signifies a vote for candidate B. Our hypothesis is that all potential paths connecting the starting point $(0, 0)$ to the final result at $(2m, 2k)$, $-m \leqslant k \leqslant m$, have an equal probability of occurrence. We will make use of the reflection principle.

(a) If A has received two votes more than B, the possible trajectories of the random walk correspond to paths from $(0, 0)$ to $(2m, 2)$. Since each of these paths consists of $m-1$ "down" steps and $m+1$ "up" steps, the total number is $\binom{2m}{m+1}$.
If candidate A must always be in the lead (which means that the number of votes for A is always greater than the number of votes for B), then the random walk must always remain strictly positive. In other words, $X_n > 0$ for all $1 \leqslant n \leqslant 2m$. The number of valid paths from the point $(0, 0)$ to $(2m, 2)$ that meet this criterion is equal to the number of paths with the same property from the point $(1, 1)$ to $(2m, 2)$. This number is given by $\binom{2m-1}{m}$ minus the number of paths from $(1, 1)$ to $(2m, 2)$ that touch or cross the horizontal n-axis. By applying the reflection principle, we can find that the latter quantity is equivalent to the number of paths from $(1, -1)$ to $(2m, 2)$, which is $\binom{2m-1}{m+1}$.
Thus, the probability that candidate A has consistently been in the lead during the counting of the votes, given that A ends up with two more votes than B, is:

$$\frac{\binom{2m-1}{m} - \binom{2m-1}{m+1}}{\binom{2m}{m+1}} = \frac{1}{m}.$$

(b) Reasoning as in part (a), we deduce that if A has received $2r$ votes more than B, then the probability that during the counting of the votes A has always been in the lead is:

$$\frac{\binom{2m-1}{m+r-1} - \binom{2m-1}{m+r}}{\binom{2m}{m+r}} = \frac{r}{m}.$$

(c) The probability that one of the candidates is always in the lead during the counting of the votes is the probability that a symmetric random walk does not return to the origin within the first $2m$ steps, which is also equal to:

$$\mathbb{P}(X_{2m} = 0) = \binom{2m}{m}\frac{1}{2^{2m}}.$$

(d) By applying the result from Problem **4(a)**, we find that the total number of trajectories of a symmetric random walk that satisfies $X_1 \geqslant 0, \ldots, X_{2m} \geqslant 0$ (where candidate A has never fallen below candidate B) is given by $\binom{2m}{m}$.

On the other hand, the number of these trajectories that conclude with an equal number of votes for both candidates corresponds to the number of paths from $(0,0)$ to $(2m,0)$ that do not touch the horizontal line $y = -1$. Utilizing the reflection principle, we can determine that this number is represented by the Catalan number:

$$C_m = \binom{2m}{m} - \binom{2m}{m+1} = \frac{1}{m+1}\binom{2m}{m}.$$

Consequently, if candidate A has never been below candidate B, the probability that the counting concludes with an equal number of votes for both candidates is given by the ratio:

$$\frac{\frac{1}{m+1}\binom{2m}{m}}{\binom{2m}{m}} = \frac{1}{m+1}.$$

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7. A vehicle is in the plane at position $(0,0)$. At every instant it moves with equal probability either to the North or to the East. We know that at instant $2n$ the vehicle is at position (n,n). Find the probability that its has not passed through any point of the form (i,i) for $0 < i < n$.

Solution

By performing a rotation of $\pi/4$, we can reformulate the problem as finding the probability that the first return to 0 of a random walk $\{X_k \colon k \geqslant 1\}$ occurs at time $2n$, with the condition that $X_{2n} = 0$. Thus, if we let Z represent the time of this first return to 0, we have the following relationship:

$$\mathbb{P}(Z = 2n \mid X_{2n} = 0) = \frac{\mathbb{P}(Z = 2n, X_{2n} = 0)}{\mathbb{P}(X_{2n} = 0)} = \frac{\mathbb{P}(Z = 2n)}{\mathbb{P}(X_{2n} = 0)} = \frac{1}{2n-1}, \tag{2}$$

where we have used the known property that $\mathbb{P}(Z = 2n) = \mathbb{P}(X_{2n} = 0)/(2n-1)$.

Remark. Another way to obtain this result, by using counting arguments, is as follows.

We want to calculate the probability that, given $X_{2n} = 0$, the trajectory of the symmetric random walk X_n for $1 \leqslant n \leqslant 2n$, corresponds to a path starting at $(0,0)$, ending at $(2n,0)$, and not touching the line $y = 0$.

On one hand, we know that the number of paths from $(0,0)$ to $(2n,0)$ which do not touch the line $y = -1$ is the Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$ (see Problem **6(d)**). On the other hand, the number of paths from $(0,0)$ to $(2n,0)$ which are always positive equals the number of paths from $(1,1)$ to $(2n-1,1)$ not touching the line $y = 0$, which is the Catalan number C_{n-1}. Thus the fraction of these paths is

$$\frac{\frac{1}{n}\binom{2n-2}{n-1}}{\binom{2n}{n}} = \frac{1}{2(2n-1)}.$$

By adding now the paths from $(0,0)$ to $(2n,0)$ which are always negative, the same number, we get the probability

$$\frac{1}{2n-1},$$

as in (2)

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8. At each time unit, a system is busy with probability $p = 1/2$ and idle with probability $q = 1-p$. At time k, the performance level of the system, denoted as X_k, is defined as the difference between the number of busy times and the number of idle times.

(a) What is the probability that the performance levels are 1, 2, and 3 at times 5, 8, and 15, respectively?

(b) What is the probability that the performance remains positive for k consecutive time units if the performance starts at 1?
[Hint: You can use that if $\{Z_n : n \geqslant 0\}$ is a symmetric random walk, then $\mathbb{P}(Z_1 \neq 0, \ldots, Z_{2m} \neq 0) = \mathbb{P}(Z_{2m} = 0)$.]

Solution

The sequence $\{X_k : k \geqslant 0\}$ is a symmetric random walk.

(a) We have

$$\begin{aligned}
\mathbb{P}(X_5 = 1, X_8 = 2, X_{15} = 3) &= \mathbb{P}(X_5 = 1)\,\mathbb{P}(X_8 = 2 | X_5 = 1)\,\mathbb{P}(X_{15} = 3 | X_5 = 1, X_8 = 2) \\
&= \mathbb{P}(X_5 = 1)\,\mathbb{P}(X_8 - X_5 = 1)\,\mathbb{P}(X_{15} - X_8 = 1) = \mathbb{P}(X_5 = 1)\,\mathbb{P}(X_3 = 1)\,\mathbb{P}(X_7 = 1) \\
&= \binom{5}{3}\left(\frac{1}{2}\right)^5 \binom{3}{2}\left(\frac{1}{2}\right)^3 \binom{7}{4}\left(\frac{1}{2}\right)^7 = \frac{525}{16384} \approx 0.03204.
\end{aligned}$$

(b) The probability that the performance remains positive for k consecutive time units if the performance starts at 1 can be expressed as $p = \mathbb{P}(X_{n+1} > 0, \ldots, X_{n+k} > 0 \,|\, X_n = 1)$. This coincides with the probability that, starting at zero, the random walk does not touch or cross the axis $y = -1$ during the first k steps:

$$p = \mathbb{P}(X_1 > -1, \ldots, X_k > -1).$$

By setting a new random walk X'_n defined as $X'_n = X_{n+1} + 1$, the above probility can be written as

$$\begin{aligned}
p &= \mathbb{P}(X'_2 > 0, \ldots, X'_k > 0, X'_{k+1} > 0 \,|\, X'_1 = 1) \\
&= 2\,\mathbb{P}(X'_1 > 0, X'_2 > 0, \ldots, X'_k > 0, X'_{k+1} > 0) = \mathbb{P}(X'_1 \neq 0, X'_2 \neq 0, \ldots, X'_k \neq 0, X'_{k+1} \neq 0).
\end{aligned}$$

We now use that, for $k = 2m$,

$$\mathbb{P}(X'_1 \neq 0, \ldots, X'_{2m} \neq 0) = \mathbb{P}(X'_{2m} = 0) = \binom{2m}{m}\left(\frac{1}{2}\right)^{2m}.$$

Moreover, if $k = 2m$ is even, then the event $\{X'_{2m+1} \neq 0\}$ happens with probability 1. Hence

$$\mathbb{P}(X'_1 \neq 0, \ldots, X'_{2m} \neq 0) = \mathbb{P}(X'_1 \neq 0, \ldots, X'_{2m} \neq 0, X'_{2m+1} \neq 0),$$

and so, if $k = 2m$,

$$p = \binom{2m}{m}\left(\frac{1}{2}\right)^{2m}.$$

Finally, if $k = 2m+1$ is odd, then $k + 1 = 2(m+1)$ is even, and therefore,

$$p = \binom{2m+2}{m+1}\left(\frac{1}{2}\right)^{2m+2}.$$

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9. In a ballot, candidate A receives m votes and candidate B receives n votes, where $m > n$,

(a) Prove that the number of voting sequences beginning with a vote for B is $\binom{m+n-1}{m}$.

(b) Use a reflection argument to show that any voting sequence that starts with a vote for A and results in a tie at some point corresponds one-to-one with a voting sequence that starts with a vote for B.

(c) From (a) and (b), deduce that the probability that candidate A was always ahead of candidate B during the vote counting is $(m-n)/(m+n)$.

Solution

We use the (n, X_n)-plane representation of the trajectories of the random walk.

(a) Any voting sequence that begins with a vote for candidate B, where candidate A receives m votes and candidate B receives n votes, can be represented as a path from the point $(1, -1)$ to the point $(m+n, m-n)$. This path has a length of $m+n-1$ and consists of m "up" steps and $n-1$ "down" steps. Therefore, the number of voting sequences starting with a vote for B is given by $\binom{m+n-1}{m}$.

(b) Any voting sequence that starts with a vote for candidate A, where A receives m votes and B receives n votes, corresponds to a path U from the point $(0, 0)$ to the point $(m+n, m-n)$, and this path begins with an initial step from $(0, 0)$ to $(1, 1)$. Additionally, if a tie occurs at any point during the voting sequence, the path U must touch or cross the horizontal n-axis for the first time at the point $(a, 0)$ for some $a > 0$.

By reflecting the segment of path U that goes from $(0, 0)$ to $(a, 0)$, we obtain a new path U' that travels from $(0, 0)$ to $(m+n, m-n)$, with its first step moving "down". Similarly, since $m-n > 0$, a path U' traveling from $(0, 0)$ to $(m+n, m-n)$ with its first step "down" corresponds to a path U that starts from $(0, 0)$ and leads to $(m+n, m-n)$ with its first step "up", which touches or crosses the n-axis at some point. Therefore, we conclude that every voting sequence that begins with a vote for A and has a tie at some point corresponds one-to-one to a voting sequence that starts with a vote for B.

(c) Let N_T be the total number of vote sequences in which candidate A receives m votes and candidate B receives n votes. Therefore, we have $N_T = \binom{m+n}{m}$. Let N'_A represent the number of these sequences that start with a vote for A and include a tie at some point. Additionally, let N_B denote the number of sequences that begin with a vote for B. Based on the results from parts (a) and (b), we find that $N'_A = N_B = \binom{m+n-1}{m}$.

Consequently, the number of vote sequences in which candidate A consistently leads candidate B, given that A receives m votes and B receives n votes, is given by $N_T - N'_A - N_B$. We can conclude that the conditional probability p that candidate A was always ahead of candidate B throughout the vote counting process, assuming A scores m votes and B scores n votes, is given by the expression.

$$p = \frac{N_T - N'_A - N_B}{N_T} = \frac{\binom{m+n}{m} - 2\binom{m+n-1}{m}}{\binom{m+n}{m}} = 1 - 2\,\frac{\frac{(m+n-1)!}{m!(n-1)!}}{\frac{(m+n)!}{m!n!}} = 1 - 2\,\frac{n}{m+n} = \frac{m-n}{m+n}.$$

Remark. Alternatively, the above probability, p, can also be derived in the following manner.

The number of voting sequences in which candidate A is always ahead of candidate B is equal to the number of sequences that start with a vote for A minus the number of sequences that start with a vote for A and experience a tie at some point. Utilising the result proved in part (b), we can count this as the difference between the number of paths from $(1, 1)$ to $(m+n, m-n)$ and the number of paths from $(1, -1)$ to $(m+n, m-n)$. This difference equals $N_d = \binom{m+n-1}{m-1} - \binom{m+n-1}{m}$, because a path from $(1, 1)$ to $(m+n, m-n)$ consists of $m-1$ "up" steps and n "down" steps, and a path from $(1, -1)$ to $(m+n, m-n)$ consists of m "up" steps and $n-1$ "down" steps. Moreover, the total number of voting sequences where candidate A receives m votes and candidate B receives n votes is given by $N_T = \binom{m+n}{m}$. Thus, we deduce that the probability p is given by the fraction

$$p = \frac{N_d}{N_T} = \frac{\binom{m+n-1}{m-1} - \binom{m+n-1}{m}}{\binom{m+n}{m}} = \frac{\frac{(m+n-1)!(m-n)}{m!n!}}{\frac{(m+n)!}{m!n!}} = \frac{m-n}{m+n}.$$

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10. Let $\{X_n : n \geqslant 0\}$ be a symmetric random walk such that $X_0 = 0$. Prove that if a is a positive integer, then

$$\mathbb{P}(X_1 \neq 0, \ldots, X_{n-1} \neq 0, X_n = a) = \frac{a}{n}\,\mathbb{P}(X_n = a).$$

[*Hint*: consider the trajectories from $(1, 1)$ to (n, a) in the (n, X_n)-plane representation of the random walk.]

Solution

Let us consider the representation of the random walk in the (n, X_n)-plane. We have the following probability expression:

$$\mathbb{P}(X_1 \neq 0, \ldots, X_{n-1} \neq 0, X_n = a) = \frac{C}{2^n},$$

where C is the number of paths from $(0,0)$ to (n,a) that do no return to the horizontal n-axis. Since the first step of such paths goes from $(0,0)$ to $(1,1)$, we can express C as $C = C_1 - C_1'$, where C_1 is the total number of paths from $(1,1)$ to (n,a), and C_1' is the number of paths from $(1,1)$ to (n,a) that touch or cross the n-axis.

Let the point $(k,0)$, where $1 \leqslant k \leqslant n-1$, be the first location at which a trajectory counted in C_1' touches the n-axis. By reflecting the segment of the trajectory from $(1,1)$ to $(k,0)$, we obtain a new trajectory from $(1,-1)$ to (n,a), and the process can be reversed as well. Therefore, the total number of paths from $(1,-1)$ to (n,a) is equal to C_1'.

Moreover, a path from $(1,1)$ to (n,a) consists of $(n+a)/2-1$ steps $\nearrow$ (and so $(n-a)/2$ steps $\searrow$); whereas a path from $(1,-1)$ to (n,a) involves $(n+a)/2$ steps $\nearrow$ (and consequently $(n-a)/2-1$ steps $\searrow$). Therefore,

$$\begin{aligned}
C &= \binom{n-1}{(n+a)/2-1} - \binom{n-1}{(n+a)/2} \\
&= \frac{(n-1)!}{((n+a)/2-1)!((n-a)/2)!} - \frac{(n-1)!}{((n+a)/2)!((n-a)/2-1)!} \\
&= \frac{(n-1)!((n+a)/2-(n-a)/2)}{((n+a)/2)!((n-a)/2)!} = \frac{(n-1)!\,a}{((n+a)/2)!((n-a)/2)!} = \frac{a}{n}\binom{n}{(n+a)/2}.
\end{aligned}$$

The proof is concluded by observing that $\mathbb{P}(X_n = a) = \binom{n}{(n+a)/2}/2^n$, and thus,

$$\mathbb{P}(X_1 \neq 0, \ldots, X_{n-1} \neq 0, X_n = a) = \frac{C}{2^n} = \frac{a}{n}\,\mathbb{P}(X_n = a).$$